Textile/fabrics/ non-wovens	Energy	Cosmetics
• Surface-processed textiles • Smart clothes	• Fuel cells • Solar cells • Batteries • Capacitors	• Sun protection • Lipsticks • Skin creams • Tooth paste
Food and drinks	**Household**	**Sports/ outdoor**
• Package materials • Storage life sensors • Additives • Clarification of fruit juices	• Ceramic coatings for irons • Odors catalyst • Cleaner for glass, ceramic, floor, windows	• Ski wax • Antifogging of glasses/goggles • Antifouling coatings for ships/boats • Reinforced tennis rackets and balls

11.2.7 Possible harmful effects of nanoparticles

The research on the harmful impact of application of nanotechnology is also equally important and fast developing. The major concern here is that the nanoparticles have the dimensions same as that of the biological molecules such as proteins. They may easily get absorbed onto the surface of living organisms and they might enter the tissues and fluids of the body.

The adsorbing nature depends on the surface of the nanoparticle. Indeed, it is possible to deliver a drug directly to a specific cell in the body by designing the surface of a nanoparticle so that it adsorbs specifically onto the surface of the target cell.

The interaction with living systems is also affected by the dimensions of the nanoparticles. For instance, nanoparticles of a few nanometers size may reach well inside biomolecules, which is not possible for larger nanoparticles. Nanoparticles can also cross cell membranes. It is also possible for the inhaled nanoparticles to reach the blood, to reach other sites such as the liver, heart or blood cells.

Researchers are trying to understand the response of living organisms to the presence of nanoparticles of varying size, shape, chemical composition and surface characteristics.

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